SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques

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Assessment: Analytical Problem Solving

Q1. Subnetting the 192.168.20.0/24 Network for Three Labs

Given: The university has the address block 192.168.20.0/24. Lab A requires a /26 subnet, Lab B a /27 subnet, and Lab C a /28 subnet. The subnets are assigned sequentially from the beginning of the /24 block without overlapping.

Method and calculations

For IPv4, the number of host bits is 32 − prefix length. The total addresses are 2^(host bits), and the usable host addresses are total addresses − 2 because the network and broadcast addresses cannot be assigned to hosts.

Lab	Prefix / Subnet mask	Block size	Network address	Valid host range	Broadcast address	Usable hosts
Lab A	/26 — 255.255.255.192	64	192.168.20.0	192.168.20.1 – 192.168.20.62	192.168.20.63	2^6 − 2 = 62
Lab B	/27 — 255.255.255.224	32	192.168.20.64	192.168.20.65 – 192.168.20.94	192.168.20.95	2^5 − 2 = 30
Lab C	/28 — 255.255.255.240	16	192.168.20.96	192.168.20.97 – 192.168.20.110	192.168.20.111	2^4 − 2 = 14

Explanation: A /26 subnet contains 64 addresses, a /27 contains 32 addresses, and a /28 contains 16 addresses. The next subnet begins at the first address after the previous subnet's broadcast address. Therefore, the three assignments do not overlap.

Final result: Lab A has 62 usable host addresses, Lab B has 30, and Lab C has 14. The address range 192.168.20.112–192.168.20.255 remains unassigned by these three lab allocations.

Q2. Identifying the Subnet for Two Systems

Given: System 1 is 192.168.1.130 and System 2 is 192.168.1.190. Both use subnet mask 255.255.255.192, which is /26.

Method and calculations

A /26 mask leaves 6 host bits. The block size in the last octet is 256 − 192 = 64. Therefore, the subnet ranges are 0–63, 64–127, 128–191, and 192–255.

System	IP address	Subnet range	Network address	Valid host range	Broadcast address	Result
System 1	192.168.1.130	192.168.1.128–192.168.1.191	192.168.1.128	192.168.1.129–192.168.1.190	192.168.1.191	Valid host address
System 2	192.168.1.190	192.168.1.128–192.168.1.191	192.168.1.128	192.168.1.129–192.168.1.190	192.168.1.191	Valid host address

Usable host calculation: 32 − 26 = 6 host bits. Total addresses = 2^6 = 64; usable hosts = 64 − 2 = 62 per subnet.

Final result: Both IP addresses fall within the same subnet, 192.168.1.128/26. System 1 (192.168.1.130) and System 2 (192.168.1.190) are both valid host addresses. Each subnet provides 62 usable host addresses.

Q3. Department Subnet Allocation in 172.16.0.0/16

Given: The organization has 172.16.0.0/16. The required subnet sizes are HR /25, Sales /26, IT /27, and Admin /28. The subnets are allocated sequentially from the start of the address block.

Method and calculations

The host count is calculated using 2^(32 − prefix length) total addresses and subtracting 2 for the IPv4 network and broadcast addresses. Each next subnet starts at the next available boundary after the preceding subnet.

Department	Prefix / Subnet mask	Network address	Valid host range	Broadcast address	Usable hosts
HR	/25 — 255.255.255.128	172.16.0.0	172.16.0.1 – 172.16.0.126	172.16.0.127	2^7 − 2 = 126
Sales	/26 — 255.255.255.192	172.16.0.128	172.16.0.129 – 172.16.0.190	172.16.0.191	2^6 − 2 = 62
IT	/27 — 255.255.255.224	172.16.0.192	172.16.0.193 – 172.16.0.222	172.16.0.223	2^5 − 2 = 30
Admin	/28 — 255.255.255.240	172.16.0.224	172.16.0.225 – 172.16.0.238	172.16.0.239	2^4 − 2 = 14

Total usable addresses allocated = 126 + 62 + 30 + 14 = 232.

Remaining unused range: 172.16.0.240 through 172.16.255.255, inclusive, assuming only the four listed subnets are allocated. This range includes the unused addresses at the end of the 172.16.0.0/24 portion and all subsequent addresses within the original /16 block.

Final result: HR receives 172.16.0.0/25, Sales receives 172.16.0.128/26, IT receives 172.16.0.192/27, and Admin receives 172.16.0.224/28. These allocations are contiguous and non-overlapping.

Q4. Subnetting 172.16.10.0/24 Using /27

Given: The original network is 172.16.10.0/24 and the subnet mask is /27 (255.255.255.224). The address to identify is 172.16.10.78; we must also check 172.16.10.95.

Method and calculations

- Bits borrowed for subnetting = 27 − 24 = 3 bits.
- Number of subnets = 2^3 = 8.
- Host bits remaining = 32 − 27 = 5 bits.
- Total addresses per subnet = 2^5 = 32; usable host addresses = 32 − 2 = 30.
- Block size in the last octet = 256 − 224 = 32.

The eight /27 subnet network addresses within 172.16.10.0/24 are .0, .32, .64, .96, .128, .160, .192, and .224.

Item	Result
Subnet containing 172.16.10.78	172.16.10.64/27 (range .64–.95)
Network address	172.16.10.64
Valid host range	172.16.10.65 – 172.16.10.94
Broadcast address	172.16.10.95
Does 172.16.10.95 fall in the same subnet range?	Yes, it is within the .64–.95 subnet block, but it is the broadcast address and cannot be assigned to a host.

Final result: The /24 network is divided into 8 subnets, each with 30 usable host addresses. The IP 172.16.10.78 belongs to 172.16.10.64/27. The address 172.16.10.95 is in that same subnet block but is its broadcast address, not a valid host address.

Q5. IPv6 Address Compression, Expansion, and /64 Prefix

Given IPv6 address: 2001:0db8:0000:0000:0000:ff00:0042:8329. The prefix length is /64.

Method and calculations

IPv6 addresses contain eight groups (hextets) of four hexadecimal digits, separated by colons. Leading zeros in a hextet may be removed. One continuous sequence of zero hextets may be replaced by :: once in an address.

1. Compressed form: 2001:db8::ff00:42:8329

2. Expanded full form: 2001:0db8:0000:0000:0000:ff00:0042:8329

3. Network prefix with /64: 2001:0db8:0000:0000::/64 (equivalently, 2001:db8:0:0::/64). The first 64 bits—the first four hextets—identify the network prefix.

4. Host/interface identifier bits = 128 − 64 = 64 bits.

5. Number of possible addresses in the /64 prefix = 2^64 = 18,446,744,073,709,551,616 addresses.

IPv6 interpretation: Unlike traditional IPv4 subnet calculations, IPv6 does not use a broadcast address, and the address count should not be reduced by two. The 2^64 figure is the total number of addresses in this /64 prefix.

Final result: The shortest compressed address is 2001:db8::ff00:42:8329, its expanded form matches the original address, the /64 network prefix is 2001:db8:0:0::/64, and it contains 2^64 possible IPv6 addresses.

Summary of Final Answers

Question	Key result
Q1	Lab A: 192.168.20.0/26 (62 hosts); Lab B: 192.168.20.64/27 (30 hosts); Lab C: 192.168.20.96/28 (14 hosts).
Q2	Both systems are in 192.168.1.128/26; valid host range .129–.190; broadcast .191; 62 usable hosts.
Q3	HR: 172.16.0.0/25; Sales: 172.16.0.128/26; IT: 172.16.0.192/27; Admin: 172.16.0.224/28; unused range begins at 172.16.0.240.
Q4	8 subnets, 30 usable hosts each; .78 belongs to 172.16.10.64/27; .95 is that subnet's broadcast address.
Q5	Compressed: 2001:db8::ff00:42:8329; prefix: 2001:db8:0:0::/64; total addresses: 2^64.

Note: The subnet assignments in Questions 1 and 3 use sequential, non-overlapping allocation because no specific starting subnet was prescribed for each lab or department.